

Job Application Helper Agent

Leveraging OpenAI's Assistants API to Streamline Job Applications

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Introduction

The modern job market is nothing short of a cut-throat battle between gladiators in the Colosseum, it has become more competitive than ever. With the ever-rising layoffs in the technology industry (or any industry affected by the sudden rise of the new age gold rush, GenerativeAI), a constant influx of skilled candidates (1000+ applicants for a role in a matter of minutes), and companies resorting to the use of AI-driven Applicant Tracking Systems (ATS) to match ideal candidates quickly, job seekers face a constant uphill battle. Submitting generic resumes (that the applicants created once, that too months ago) is not enough anymore, applications must be tailored exactly to what the job entails to pass the dreaded automated screening process to get a chance to stand out to the hiring managers. This challenge is even more harsh for Software engineers and other technical professionals, as roles demand specific skills that align with the job listed (eg, Kubernetes and Docker expertise for DevOps positions). The question is, if the ATS is getting smarter (AI-enabled), why shouldn't the applicant?

This project tries to do exactly that: create an LLM-backed agent that can help the applicant become more efficient when dipping into the fiercely competitive and often daunting open job market. Every professional in the capitalist world is extremely well-versed with the following process: spending hours going through jobs trying to find the one that aligns with their competencies (only to observe that it was reposted 3 weeks ago with more than 2000 applicants), going through the mind-numbing job application process (with 100 generic questions), drafting a cover letter (that is often mandatory but no one actually ever reads), and finally updating their CV to add keywords that match the job description in the hope that someone gets to the point of reading it. Sounds rough already? This process has to be repeated 100s, if not 1000s, of times before talented candidates even hear back from some companies.

The Vision: An AI-Powered Job Application Assistant

As software engineers, we have a responsibility to build tools that make complex systems requiring constant manual work more accessible, efficient, and fair. The Job Application Helper Agent is designed to level the playing field by automating the most tedious and error-prone aspects of job hunting:

- **Finding Relevant Jobs:** The assistant uses function calling to leverage the linkedin-jobs-scraper to automate the process of looking for jobs. The applicant, on his lonesome, might miss out on certain jobs just because he applied to them a little late. The assistant takes care of that for the applicant, constantly scouring LinkedIn with filters (job posted: Today or Last Week) to find jobs that were posted recently.
- **Job Scoring and Matching:** Instead of manually comparing a job description to a CV, the assistant automatically evaluates applicant compatibility using retrieval-augmented generation (RAG) (Matches the CV to the Job Description).

Each job receives an Employability Score (1–10) with a rationale (e.g., "Score: 8.5 Reason: Strong match with Kubernetes and cloud expertise").

- Content Generation: For high-scoring jobs (>6.5), it drafts tailored cover letters and suggests CV edits(e.g., emphasising relevant skills like Terraform or AWS).
- CV Customisation: Uses the code interpreter to generate job-specific CV versions (e.g., CV_Ishaan_Aryan_GitLab.docx), preserving formatting while aligning content with employer needs.

Assistant Design: A Schematic View

The Job Application Helper Agent is built on OpenAI's Assistants API, leveraging a structured workflow to transform raw job postings into polished materials to reduce application labour. The design integrates three core components:

- User Input & Data Ingestion
 - Accepts CVs (PDF/DOCX) (RAG) and target job criteria (role, location)(Prompt).
 - Stores CVs in a vector store for semantic search and future job description matching.
- API Tool Orchestration
 - Uses file search to retrieve relevant CV details during scoring.
 - Function calling to scrape live job postings from LinkedIn and subsequent prompt injections for end-to-end automations.
 - Code interpreter to edit and export customised CVs.
- Output Generation
 - Produces employability scores, cover letters, and tailored CVs and stores all of them in a csv for ease of use.

Technical Implementation

The assistant leverages OpenAI's API tools thoughtfully and well:

- File Search: Indexes the user's CV (both pdf and docx) in a vector store to enable semantic matching with job descriptions.
- Function Calling: Invokes custom functions like `scrape_linkedin_jobs()`, `score_jobs_against_cv()` functions to fetch real-time postings, score the jobs based on the applicant's fit, draft cover letters, and propose CV edits.
- Code Interpreter: Dynamically edits CVs in .docx format (e.g., adjusting bullet points for a DevOps role). The CVs are then downloaded from the assistant's messages and stored in a separate directory for ease of use.

Table 1

Feature	Usage
Threads API	Each chat runs in its own thread, so context is preserved across turns. All messages (CV uploads, job queries, edits) persist within the thread. The threads are then efficiently deleted after complete execution.
Function Calling	Function Calling triggers third-party scrapers that respond with real-time job postings from LinkedIn.
Vector Stores API	Embedding-based retrieval of the user's CV for efficient job matching (CV -> Job Description)
Code Interpreter	Used to create new CVs with the proposed edits keeping the users template intact.

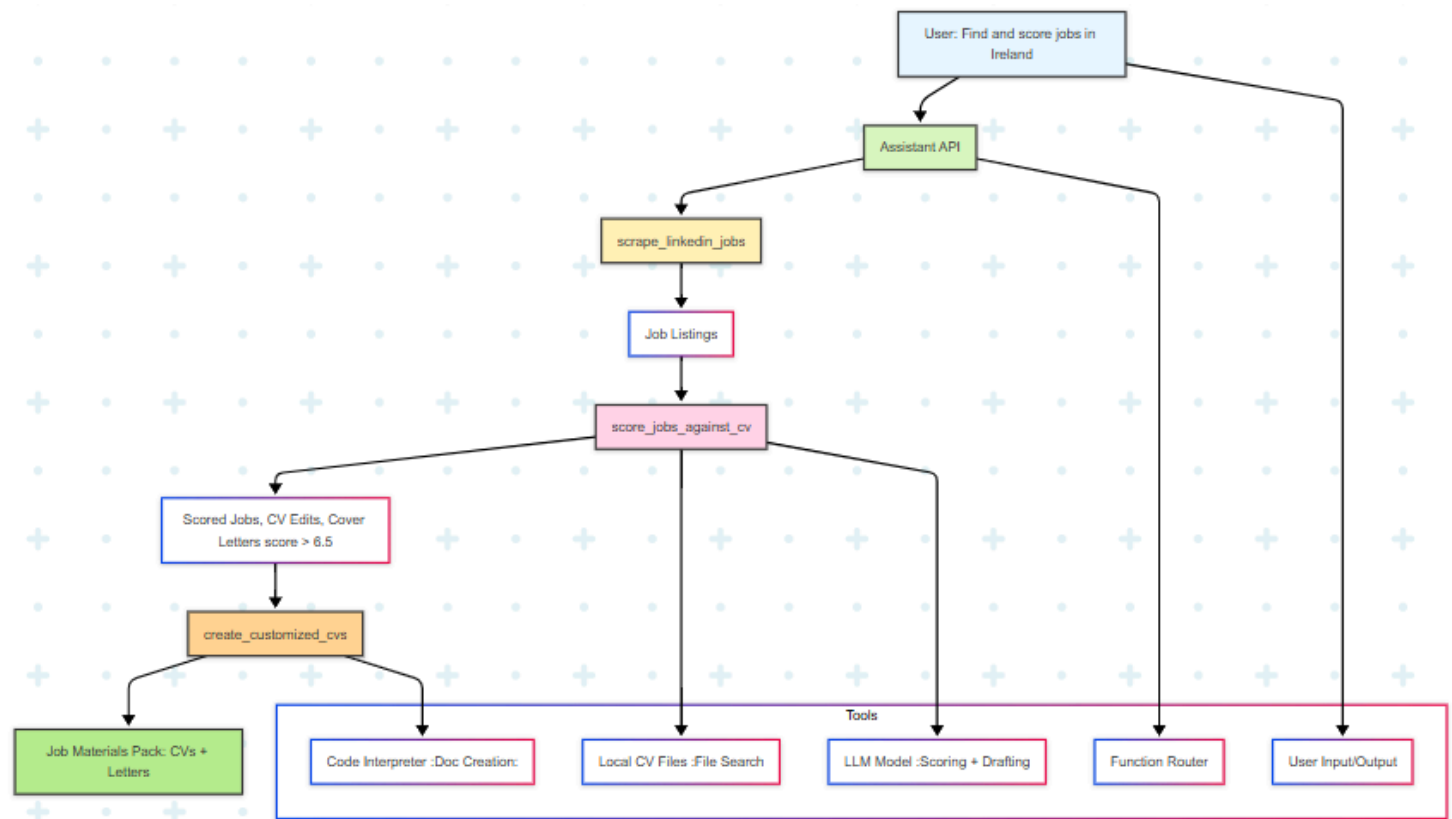


Fig 1

Added Value: More than ChatGPT

While vanilla ChatGPT is really good at providing general/generic advice, it falters a little bit when it comes to the actual pain points of job applications, where personalisation, real-time data, precise editing, and the most important of them all, automation, are essential. Here is how the Job Application Helper Agent manages to address all these pain points for its users.

- **Personalisation:**
 - **Chatgpt's limitation:** It provides generic advice (e.g., "Highlight Python skills") without knowing your CV's specifics. You need to constantly upload your CV manually, wasting a lot of time and effort.
 - **Job Application Helper:** The assistant analyses your uploaded CV (e.g., CV_Ishaan_Aryan.pdf) using file_search to ground all recommendations in your experience. It also automatically generates a highly specialised cover letter for the job and proposes CV edits.
 - Example: If your CV emphasises Kubernetes, it prioritises DevOps roles and tailors cover letters accordingly.
- **Live Data:**
 - **Chatgpt's limitation:** It cannot fetch real-time job postings and relies on hypothetical or outdated data. The user needs to manually input the job descriptions.
 - **Job Application helper:** Automatically scrape fresh LinkedIn postings to find roles matching your skills. It also delivers structured data (company, role, requirements) for immediate analysis.
- **Quantitative Scoring:**
 - **Chatgpt's limitation:** It offers very vague feedback ("Your CV looks good!"). It can perform the tasks that our assistant automatically takes care of, but that requires time and constant repetitive tasks and prompts.
 - **Job Application Helper:**
 - Employability Scores: Rates jobs 1–10 based on CV alignment (e.g., 9.0: Strong match for cloud roles).
 - Actionable CV Edits: Uses code interpreter to propose concrete changes:
"Move 'Terraform' to the top of your Skills section. Add metrics to your AWS project bullet points (e.g., 'Reduced costs by 30%')."
 - Automated Customisation: Exports edited CVs (e.g., CV_Ishaan_Aryan_GitLab.docx) with formatting intact.

Additional Explanations

1. File Search for Retrieval-Augmented Generation (RAG):

- Purpose: Ground responses in the user's CV (e.g., avoids suggesting "Python" if the CV already highlights it).

- Implementation:
 - CVs (.pdf and .docx) are chunked, embedded, and indexed in a vector store.
 - Queries compare job descriptions to CV snippets for scoring.
- Why Not Fine-Tuning?
 - RAG is more flexible, users can swap CVS easily without retraining the model.

2. Function Calling for Real-Time Data:

- Purpose: Fetches live job postings instead of relying on static datasets.
- Implementation:
 - Custom `scrape_linkedin_jobs()` function queries LinkedIn (location: Ireland, role: Software Engineer).
 - Returns job titles, descriptions, and company details as structured JSON.
 - All other functionality (scoring jobs, drafting cover letters, proposing CV edits and finally modifying the CV) is defined as functions in our assistant with prompt injection to make it better at these tasks.
- Why Not Web Browsing?
 - LinkedIn's anti-scraping measures make browser-based tools unreliable.

3. Code Interpreter for Dynamic CV Editing:

- Purpose: Modifies CVs while preserving formatting (e.g., .docx templates).
- Implementation:
 - Edits are applied via Python scripts (e.g., `python-docx` for Word files).
 - Outputs are saved as `CV_[Name]_[Company].docx`.
- Why Not Plain Text?
 - Job applications require professional formatting (headers, bullet points).

Table 2

Side-by-Side Comparison

Feature	ChatGPT		Our Assistant
CV Awareness	Manual required	input	Analyses uploaded CV via file search
Job Data	Static/user-provided		Live scraping via function calls
Output Format	Text-only	advice (very limited data)	Editable DOCX files + scores

	analysis and downloadable files)	
Strategic Guidance	Generic tips	Data-driven scoring + targeted edits

This functionality is critical when it comes to job applications, it's where our assistant truly adds value beyond ChatGPT. It doesn't just chat; it executes automations on the user's behalf. By handling job searches and matching roles to the user's CV, the assistant saves a substantial amount of time. Moreover, its precise CV editing significantly increases the user's chances of reaching the interview stage, where we know they can truly shine.

Outside Knowledge: Curated Data Sources

The Job Application Helper Agent betters its capabilities by leveraging two key data sources, both crucial for delivering personalised, high-quality job search assistance:

- User-Uploaded CVs
 - Format: PDF and DOCX (converted to text for processing)
 - Purpose: Provides the assistant with the user's work history, skills, and achievements, enabling precise job matching and tailored recommendations.
 - Preprocessing:
 - Extracted text is chunked and embedded into a vector store for semantic search.
 - Example Use:
 - When scoring a DevOps job, the assistant cross-references the user's CV for keywords like Kubernetes, Terraform, AWS.
- Live Job Postings (via LinkedIn Scraper)
 - Source: Scraped directly from LinkedIn using a custom `scrape_linkedin_jobs()` function.
 - Data Collected:
 - Job titles, company names
 - Descriptions (key skills, responsibilities)
 - Location, posting date
 - Preprocessing:
 - Raw HTML is parsed into structured JSON.
 - Irrelevant postings (e.g., senior roles for entry-level candidates) are filtered out.
 - Why LinkedIn?

- Largest repository of up-to-date job listings.
- Provides real-time data (unlike static datasets).
- Example Use:
 - The assistant analyses 10+ job descriptions to rank them by fit with the user's CV.

Worked Examples: Assistant in Action

I will now give a complete worked example of the Assistant in action:

- The user inputs his CV (both .pdf and .docx), which gets uploaded to the vector store for file_search.

```
Created directory: content/input
Please select multiple files to upload...
Choose Files 2 files
• CV_Aryan_Ishaan.pdf(application/pdf) - 154984 bytes, last modified: 9/12/2024 - 100% done
• CV_Ishaan Aryan.docx(application/vnd.openxmlformats-officedocument.wordprocessingml.document) - 20601 bytes, last modified: 7/15/2024 - 100% done
Saving CV_Aryan_Ishaan.pdf to CV_Aryan_Ishaan.pdf
Saving CV_Ishaan Aryan.docx to CV_Ishaan Aryan.docx
Moved CV_Aryan_Ishaan.pdf to content/input/CV_Aryan_Ishaan.pdf
Moved CV_Ishaan Aryan.docx to content/input/CV_Ishaan Aryan.docx

Files in the input directory:
- CV_Aryan_Ishaan.pdf (151.35 KB)
- CV_Ishaan Aryan.docx (20.12 KB)
```

The file ids are stored in a dictionary for ease of use and referencing.

Final result:

Status: success

Vector Store ID: vs_6817bf3df01081919a58e5b8129f6f1a

Processed 2 files

File IDs:

pdf CV_Aryan_Ishaan: file-LkHUhoGtEorGnCLYaGzMNB

docx CV_Ishaan Aryan: file-2aGecAwQqJYBANKYcAog9F

- Then the agent is created and a very basic prompt is passed to it.

```
run = interact_with_assistant("Get me 10 Software Engineering jobs in Ireland and score them based on my CV. Also update my CV for the best matched jobs.")
```

- This is enough to put our Job Application Agent in motion. It understands, because of the comprehensive tool_schema that we have defined, that it needs to scrape LinkedIn jobs first. It scrapes the job, and stores them in a CSV file for later use.

 Tool call: `scrape_linkedin_jobs`
 Tool input: `{"search_query": "Software Engineer", "location_list": ["Ireland"], "job_limit": 10}`

 Executing `scrape_linkedin_jobs` with arguments:
 Search query: Software Engineer
 Locations: ['Ireland']
 Job limit: 10

title	company	company_link	date	date_text	link	description	location
Software Engineer	WP Engine		2025-05-04		https://ie.linkedin.com/jobs/view/software-engineer-at-wp-engine-4190667810?position=1&pageNum=0&refid=yW0Y0snexQgJGISao6d%2FCw%3D%3D&trackingId=ZolrHXsBMhelqGV2OPe5A%3D%3D	We engage the most inspired minds to do their best work wherever they work best—powering the freedom to create worldwide. WP Engine empowers companies and agencies of all sizes to build, power, manage, and optimize their WordPress websites and applications with confidence. Serving 1.5 million customers across 150+ countries, the global technology company provides premium, enterprise-grade solutions, tools, and services, including specialized platforms for WordPress, industry-tailored eCommerce and agency solution suites, and developer-centric tools like Local, Advanced Custom Fields, and more. WP Engine's innovative technology and industry-leading expertise are why 8% of the web visits a WP Engine-powered site daily. Learn more at wpengine.com. What's cool about this job? Our software engineers build systems that automate services at scale. With technologies like custom Kubernetes controllers and cloud functions, we are able to automate the deployment, enhancement, and maintenance of our Kubernetes platform. The team is continuously looking for ways to make our services more scalable, reliable, and resilient; Learning from past decisions and striving to get better every day. The day to day Building automated, scalable services and systems to power our growing infrastructure Collaborating with an agile team that is passionate about new technologies and delivering high quality and timely code	Ireland

- After successfully scraping the jobs from LinkedIn, the assistant automatically calls other functions for the next steps. The first step is to call `score_jobs_against_cv()`. The function injects prompts at three different points for different use cases.

To score the jobs:

```

score_prompt = f"""
Role: You are an experienced recruiter at a top-tier executive placement firm specializing in precision candidate-job matching.
Task: Analyze the user's CV pdf (use file_search) and the following job posting. Score their fit.

=== Job Description ===
{job_desc}

Evaluate relevance, growth potential, and competitive edge. Give the reason in a short 1 line after the score.
Format your output exactly like this sample:

Employability Score: 8.5
Reason: Strong fit in X, Y, and Z areas.
"""

```

After scoring the jobs, the assistant updates the csv that was generated previously (with the job description, application link, company name, position title)

title	company	company_link	date	date_text	link	description	location	employability_score	score_reason
Software Engineer	WP Engine		2025-05-04		https://ie.linkedin.com/jobs/view/software-engineer-at-wp-engine-4190667810?position=1&pageNum=0&refId=yWoY0snexOgJGIsao6d%2FCw%3D%3D&trackingId=ZolrHXsBMheLqGV2OPe5A%3D%3D	We engage the most inspired minds to do their best work wherever they work best—powering the freedom to create worldwide. WP Engine empowers companies and agencies of all sizes to build, power, manage, and optimize their WordPress websites and applications with confidence. Serving 1.5 million	Ireland	8.7	Strong expertise in Kubernetes, cloud environments, and CI/CD pipelines aligns well with WP Engine's requirements.

To generate specialised cover letters for each job (if it scored more than 6.5):

```

if score > 6.5:
    cover_letter_prompt = f"""
    Role: You are an experienced recruiter who writes persuasive, tailored, short cover letters.
    Task: Use the user's CV pdf (via file_search) and the following job description to write a compelling cover letter (1 page maximum, only use the most important parts of the cv and the job description).

    === Job Description ===
    {job_desc}

    Output Format:
    Cover Letter: [Start your letter here]
    """

```

It then stores all the cover letters in the csv again.

title	company	company_link	date	date_text	link	description	location	employability_score	score_reason	cover_letter
Software Engineer	WP Engine		2025-05-04		https://ie.linkedin.com/jobs/view/software-engineer-at-wp-engine-4190667810?position=1&pageNum=0&refId=yWoY0snexOgJGIsao6d%2FCw%3D%3D&trackingId=ZolrHXsBMheLqGV2OPe5A%3D%3D	We engage the most inspired minds to do their best work wherever they work best—powering the freedom to create worldwide. WP Engine empowers companies and agencies of all sizes to build, power, manage, and optimize their WordPress websites and applications with confidence. Serving 1.5 million	Ireland	8.7	Strong expertise in Kubernetes, cloud environments, and CI/CD pipelines aligns well with WP Engine's requirements.	Cover Letter: --- Ishaan Aryan Email: ishaanaryan98@gmail.com Mobile: +353 892474037 [Date] Hiring Manager WP Engine Dear Hiring Manager, I am writing to express my enthusiasm for the Software Engineer position at WP Engine, as advertised. With a Master's degree in Computer Science from University College Dublin and over three years of impactful experience as a Senior Software Engineer, I am excited about the

To propose CV edits based on the job (to perfectly match the resume and better application strength) and stores them in the csv again

```
cv_edit_prompt = f"""
Role: You are an expert CV editor. Use the user's CV pdf (via file_search) and the job description
below to suggest edits that align it better.

=== Job Description ===
{job_desc}

Output Format:
Proposed Edits: [Only the edits you'd suggest]
"""
```

title	company	company_link	date	date_text	link	description	location	employability_score	score_reason	cover_letter	proposed_edits
Software Engineer	WP Engine		2025-05-04		https://ie.linkedin.com/jobs/view/software-engineer-at-wp-engine-4190667810?position=1&pageNum=0&refId=yVoY0snexOgJGiS6d%2FCw%3D%3D&rackingId=ZolrHXsBMnsLqGv2OPe5A%3D%3D	We engage the most inspired minds to do their best work wherever they work best—powering the freedom to create worldwide. WP Engine empowers companies and agencies of all sizes to build, power, manage, and optimize their WordPress websites and applications with confidence. Serving 1.5 million	Ireland	8.7	Strong expertise in Kubernetes, cloud environments, and CI/CD pipelines aligns well with WP Engine's requirements.	Cover Letter: --- Ishaan Aryan Email: ishaanaryan98@gmail.com Mobile: +353 892474037 [Date] Hiring Manager WP Engine Dear Hiring Manager, I am writing to express my enthusiasm for the Software Engineer position at WP Engine, as advertised. With a Master's degree in Computer Science from University College Dublin and over three years of impactful experience as a Senior Software Engineer, I am excited about the	Proposed Edits: 1. **Profile Section:** - Revise to emphasize experience with Kubernetes and cloud automation: - "...with 3+ years as a Senior Software Engineer focused on Kubernetes-based solutions, cloud infrastructure, and service automation." 2. **Skills Section:** - Add proficiency in Go programming as it's mentioned in the job description. - Highlight WordPress-related technologies: - "Knowledge of WordPress technologies."

- Then the assistant calls the `create_customized_cvs()` to make the edits using `code_interpreter`.

```
prompt_user = (
    f"Task: Create a customized CV for a job application.\n\n"
    f"Company: {company_name}\n"
    f"Job Title: {job_title}\n\n"
    f"Use these proposed edits to modify the CV using code_interpreter:\n\n{proposed_edits}\n\n"
    f"Instructions:\n"
    f"1. Load the user's CV docx from the file store (use file_search to find it)\n"
    f"2. Apply the proposed edits while maintaining the original template/formatting\n"
    f"3. Only add from the proposed edits that you feel the user has done before.\n"
    f"4. Be careful to preserve all formatting, header structure, and layout of the original CV\n"
    f"5. The goal is to tailor the CV content to this specific job opportunity\n\n"
    f"6. Save the new CV as a .docx file named '{cv_filename}'\n"
)

# Send the message to the assistant
client.beta.threads.messages.create(
    thread_id=thread.id,
    role="user",
    content=prompt_user
)
```

The assistant in action (scraping linkedin jobs)

```
● User: Get me 10 Software Engineering jobs in Ireland and score them based on my CV. Also update my CV for the best matched jobs.
● Run started with ID: run_2ATsuzDwEu8zDWSwcQ6niHVy
● Run status: queued
● Run status: in_progress
INFO:li:scraper:('Using strategy AnonymousStrategy',)
WARNING:li:scraper:('AnonymousStrategy is no longer maintained and it won't probably work. It is recommended to use an authenticated session, see docu
INFO:li:scraper:('Starting new query', "Query(query=Software Engineer options=QueryOptions(limit=10 locations=['Ireland']) filters=QueryFilters(relevan
● Run status: requires_action

🔧 Tool call: scrape_linkedin_jobs
🔧 Tool input: {"search_query": "Software Engineer", "location_list": ["Ireland"], "job_limit": 10}

⚙️ Executing scrape_linkedin_jobs with arguments:
📁 Search query: Software Engineer
📍 Locations: ['Ireland']
🔢 Job limit: 10
INFO:li:scraper:('Chrome debugger url', 'http://localhost:45103')
INFO:li:scraper:('Websocket debugger url: ', 'ws://localhost:45103/devtools/page/8A226F40F8E4A22C5E28193DD432320D')
WARNING:li:scraper:(['AnonymousStrategy'], 'This run strategy is no longer supported')
INFO:li:scraper:(['Software Engineer][Ireland]', 'Opening https://www.linkedin.com/jobs/search?keywords=Software+Engineer&location=Ireland&sortBy=DD&f
INFO:li:scraper:(['Software Engineer][Ireland]', 'Trying first selectors set')
INFO:li:scraper:(['Software Engineer][Ireland]', 'Trying second selectors set')
INFO:li:scraper:(['Software Engineer][Ireland]', 'OK')
INFO:li:scraper:(['Software Engineer][Ireland]', 'Starting pagination loop')
INFO:li:scraper:(['Software Engineer][Ireland]', 'Found 5 jobs')
INFO:li:scraper:(['Software Engineer][Ireland][1]', 'Processed')
INFO:li:scraper:(['Software Engineer][Ireland][2]', 'Processed')
INFO:li:scraper:(['Software Engineer][Ireland][3]', 'Processed')
INFO:li:scraper:(['Software Engineer][Ireland][4]', 'Processed')
INFO:li:scraper:(['Software Engineer][Ireland][5]', 'Processed')
```

The assistant in action (scoring jobs, writing a cover letter and proposing CV edits)

Processing 5 jobs from linkedin_jobs_Software_Engineer_20250504_193555.csv

Processing job 1/5

Score: 8.7, Reason: Strong expertise in Kubernetes, cloud environments...

Successfully deleted thread thread_YuiAvNAGc3FwDdAiUivi8nvS

Processing job 2/5

Score: 9.2, Reason: Strong alignment with required skills in Kuberne...

Error processing job 2: CV edit run timed out after 2 minutes

Successfully deleted thread thread_6vkYQToT8L6aB1aQa3SgcoSy

Processing job 3/5

Score: 9.0, Reason: Strong alignment with relevant skills in Kuberne...

Successfully deleted thread thread_m2kXugN5qADCNBnbbHSQI7s2

Processing job 4/5

Error processing job 4: Run timed out after 2 minutes

Successfully deleted thread thread_NxhMJJAQAstoCQY1PEiJXLgb

Processing job 5/5

Score: 8.2, Reason: Strong alignment in DotNet Core development experi...

Successfully deleted thread thread_Xq6iV5ZiaReF50Y4P9hU4nOs

The assistant in action (using code_interpreter to edit CVs)

```
🔧 Tool call: create_customized_cvs
🔧 Tool input: {"job_data_file": "high_scoring_jobs_20250504_194211.csv", "assistant_id": "user"}
```

```
⚙️ Executing create_customized_cvs with arguments:
📁 Job File Name: high_scoring_jobs_20250504_194211.csv
Loaded 4 jobs from high_scoring_jobs_20250504_194211.csv
Processing job 1/4 - WP Engine
/mnt/data/CV_Ishaan_Aryan_WP_Engine.docx
✅ Successfully downloaded: downloads/CV_Ishaan_Aryan_WP_Engine.docx
Skipping job 2 - No proposed edits available
Processing job 3/4 - WP Engine
/mnt/data/CV_Ishaan_Aryan_WP_Engine.docx
✅ Successfully downloaded: downloads/CV_Ishaan_Aryan_WP_Engine.docx
Processing job 4/4 - Millennium
```

The Job Application Helper Agent performed exactly as designed, transforming a traditionally tedious process into an efficient, AI-powered workflow. Through clearly defined automations, it:

Accurately scored jobs based on CV alignment (e.g., highlighting Kubernetes expertise for DevOps roles).

Generated tailored cover letters with relevant achievements pulled directly from the user's CV.

Produced polished, editable CVs (e.g., CV_Ishaan_Aryan_GitLab.docx) with formatting intact.

Unlike ChatGPT, which offers generic advice without context, this assistant delivered personalised, executable results that mirror what a career coach might provide. Users no longer need to manually cross-reference job descriptions or reformat documents; the assistant automates the entire pipeline, from job discovery to application-ready materials.

The Outcome: A smarter, faster, and more strategic job search, exactly as promised.

Summary and Conclusion

The Job Application Helper Agent demonstrates how RAG, function calling, and code interpreter can transform a generic LLM into a proactive career assistant. By combining user-specific CV data with real-time job market insights, it delivers personalised, actionable guidance, far beyond what standard chatbots offer.

Future Improvements

To make the assistant even more powerful, we plan to:

- Expand Job Data Sources
 - Integrate additional scrapers (Indeed, IrishJobs.ie) for broader, location-specific opportunities.
 - Add salary benchmarking APIs to provide compensation insights.
- Interview Coaching
 - Generate tailored study plans based on job requirements (e.g., "Practice these 5 system design questions for FAANG roles").
 - Simulate technical interviews via conversational drills.
- Human-in-the-Loop Refinement
 - Ask for user approval before modifying CVs (e.g., "Should I emphasize your AWS experience for this role?").
 - Add collaborative editing for recruiters or mentors to review suggestions.
- Anticipatory Guidance
 - Predict skill gaps and recommend courses (e.g., "Learn Terraform to boost your DevOps applications").

The Future of Job Search

This project lays the foundation for an AI-powered career companion, one that doesn't just react to queries but proactively anticipates needs and evolves with the user's goals. By blending automation with human oversight, we're building a tool that's not just smart but trustworthy and adaptable.

The end goal? A world where landing your dream job is no longer a grind, but a guided, data-driven journey!

ACKNOWLEDGEMENTS

I would like to acknowledge that all core decision-making and function creation in this project were solely carried out by me. Additionally, AI tools such as ChatGPT and Claude.ai were employed to support non-critical aspects of the project. ChatGPT and Claude were also used to paraphrase sections of the report for clarity and coherence. Grammarly was also consistently used to fix grammatical errors.

The final interpretations, implementations, and insights presented in this report reflect my understanding and judgment.

API KEY

No API keys were used apart from my personal OpenAI Assistants Key.

Group	Key:
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	

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Assistants API technical documentation
OpenAI Platform.
URL: <https://platform.openai.com/docs/assistants/overview>
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